GUARDIAN OF THE EGYPTIAN ROADS

Egypt

PRESENTED BY MOHAMED YOUNIS

Next-Generation AI License Plate Detection & OCR

Executive Summary

Deploying advanced Computer Vision to automate vehicle monitoring across Cairo's evolving road network. Aligned with Egypt Vision 2030 for smart city development.

Key Objectives

- 98%+ Detection Accuracy
- Real-time Arabic OCR Processing
- Scalable Edge Deployment
- Security & Law Enforcement Integration

The Urban Challenge

The sheer volume of Egyptian traffic requires an automated, AI-driven solution to ensure safety, streamline tolls, and manage urban growth.

10M+
Registered Vehicles

24/7
Continuous Flow

110M
Population Density

Decoding the Visual Identity

COMPLEXITY MANAGEMENT

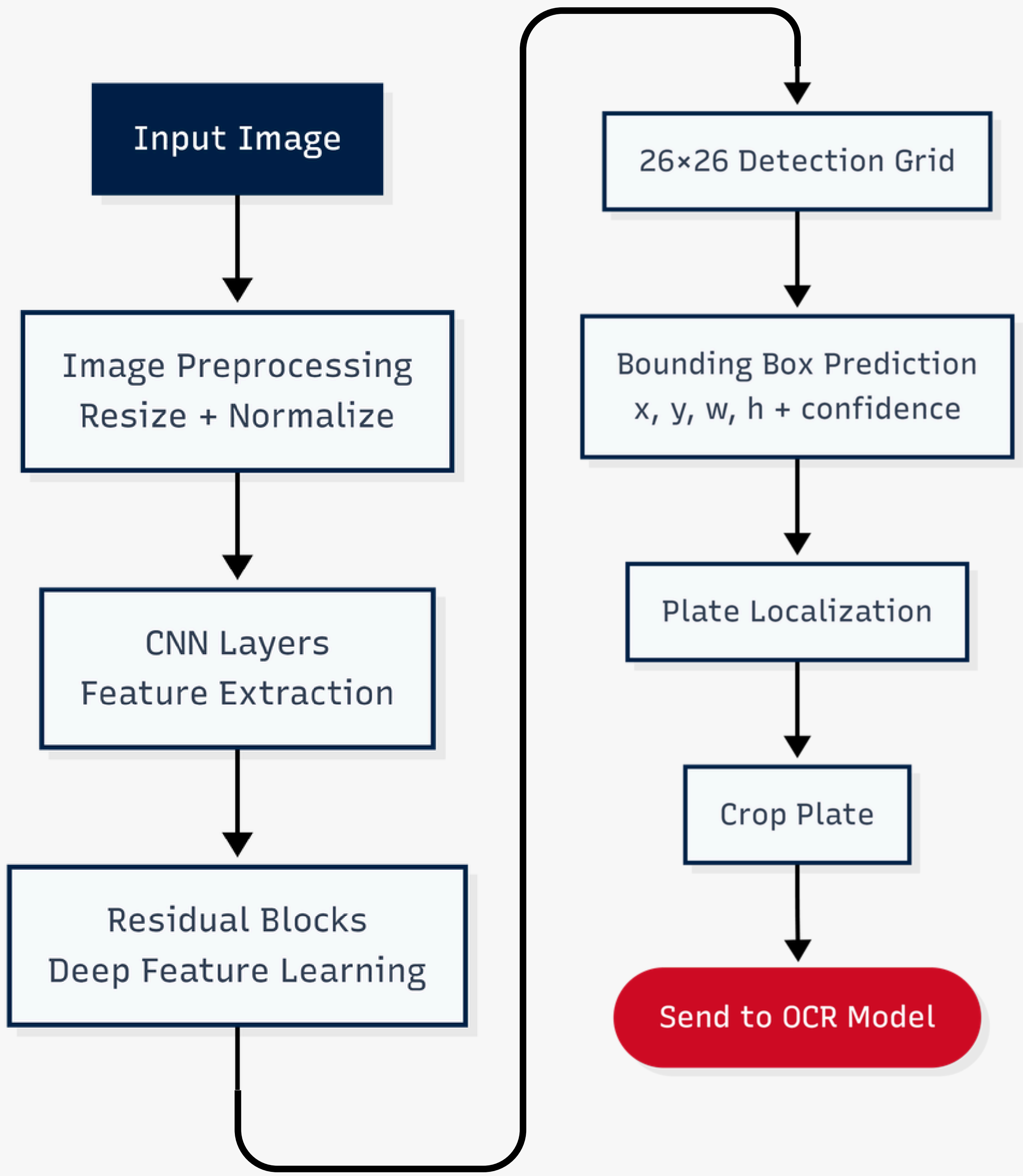
UNLIKE LATIN SCRIPTS, ARABIC PRESENTS UNIQUE CHALLENGES IN OCR INCLUDING CHARACTER LIGATURES AND DOT-BASED DIFFERENTIATION:

- RTL LOGIC: PROCESSING FROM RIGHT-TO-LEFT.
- DOT RECOGNITION: PRECISION IN DETECTING 'ب', 'ت', 'ث'.
- DUAL FORMATS: RECOGNIZING BOTH ARABIC AND ENGLISH ALPHANUMERIC CHARACTERS

Technological Framework

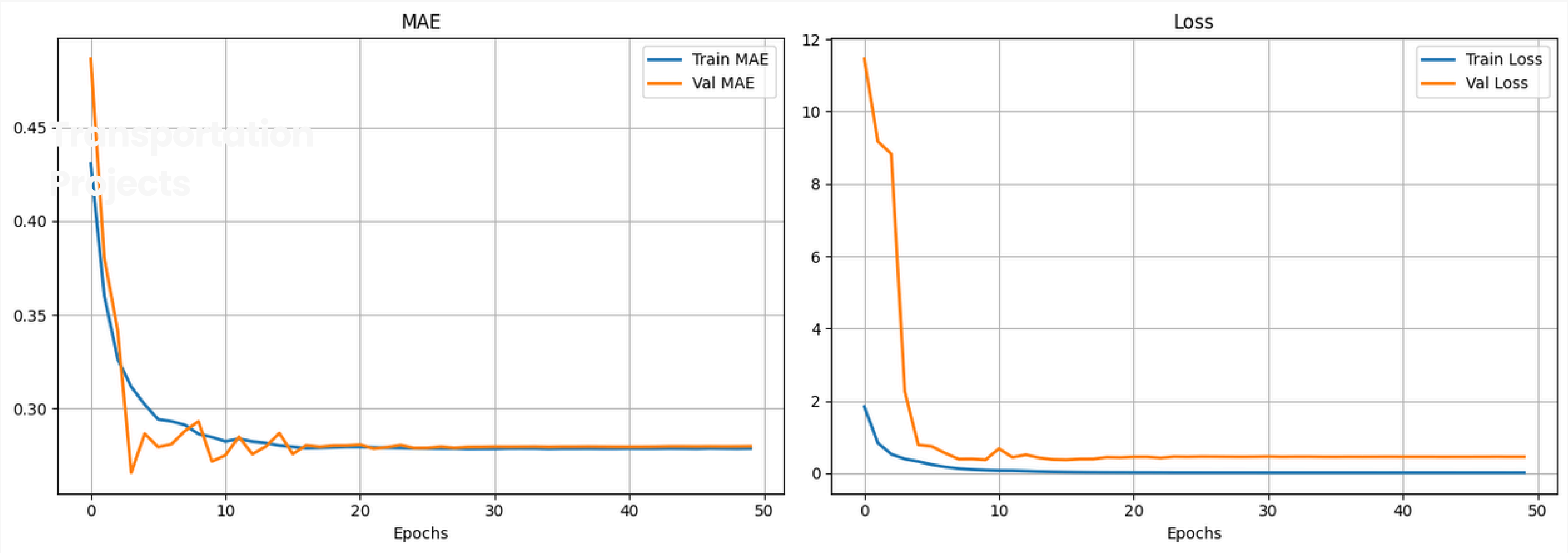
LICENSE PLATE DETECTION PIPELINE - CNN

- **Input & Pre-processing:** Resizes raw images to 416×416 for optimal real-time processing.
- **Feature Extraction (CNN):** Isolates critical visual data like edges, shapes, and plate boundaries.
- **Residual Blocks:** Uses skip connections to prevent vanishing gradients in deep layers.
- **Grid-Based Detection:** Divides the frame into a 26×26 grid to locate the plate.
- **Bounding Box Regression**: Predicts 5 metrics: [Confidence, X, Y, Width, Height].
- **ROI Output:** Crops the isolated plate area and routes it directly to the OCR engine.



Model Evaluation & Qualitative Results

LICENSE PLATE DETECTION PIPELINE - CNN



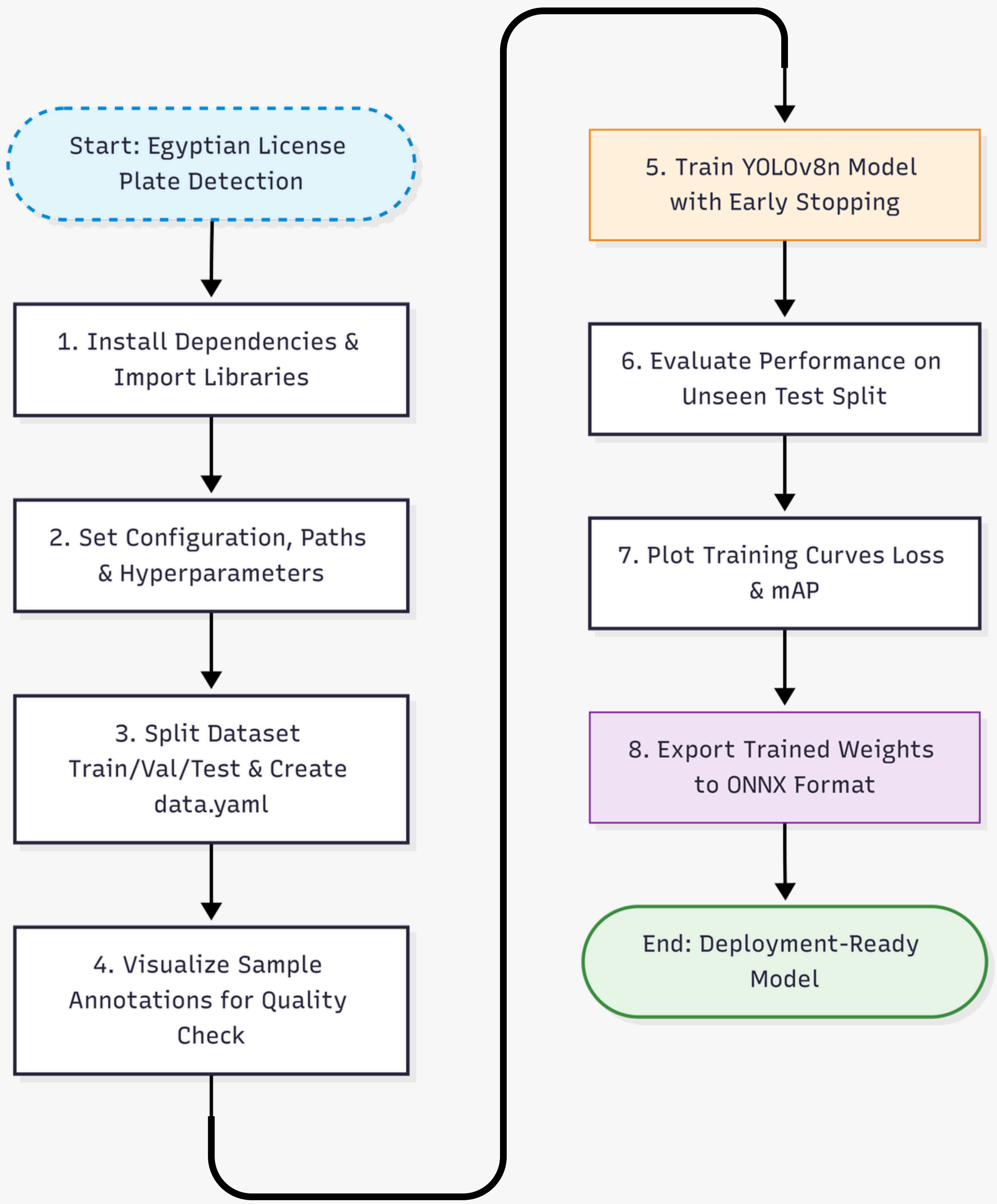
PERFORMANCE METRICS (GRAPHS)

INFERENCE SAMPLES

Technological Framework

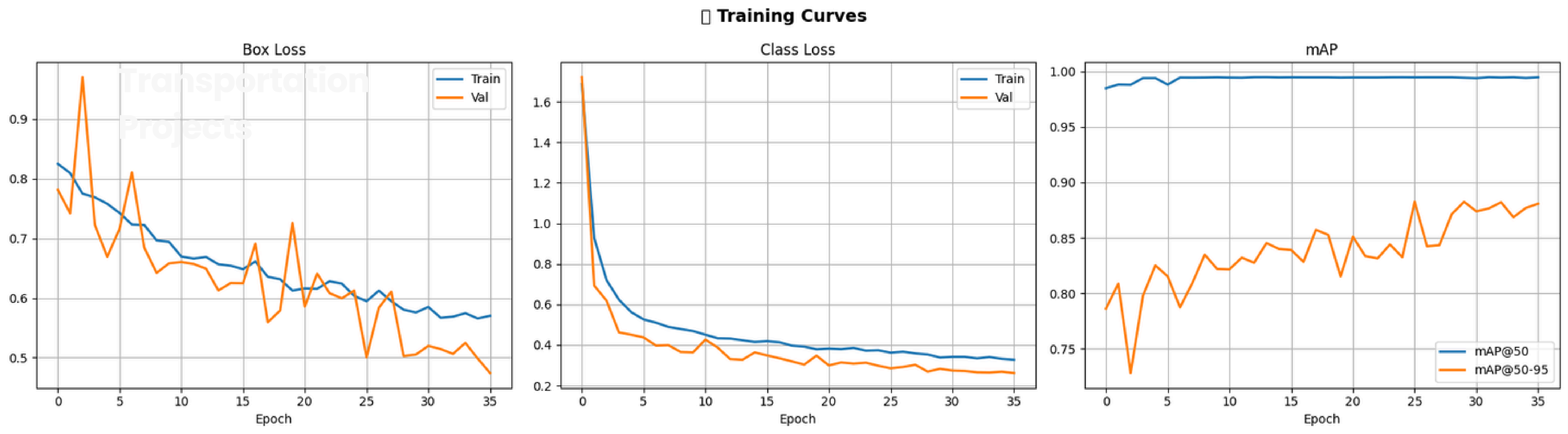
LICENSE PLATE DETECTION PIPELINE - YOLO

- **Environment Setup:** Installed ultralytics and imported core deep learning libraries.
- **Configuration:** Configured dataset paths, image size (640), batch size (16), and epochs.
- **Data Preparation:** Split dataset into Train/Val/Test and generated data.yaml.
- **Sanity Check:** Visualized sample training images to verify bounding box accuracy.
- **Model Training:** Fine-tuned the pre-trained YOLOv8n model on custom plate data.
- **Model Export:** Converted the final trained weights (best.pt) to ONNX format.



Model Evaluation & Qualitative Results

LICENSE PLATE DETECTION PIPELINE - YOLO



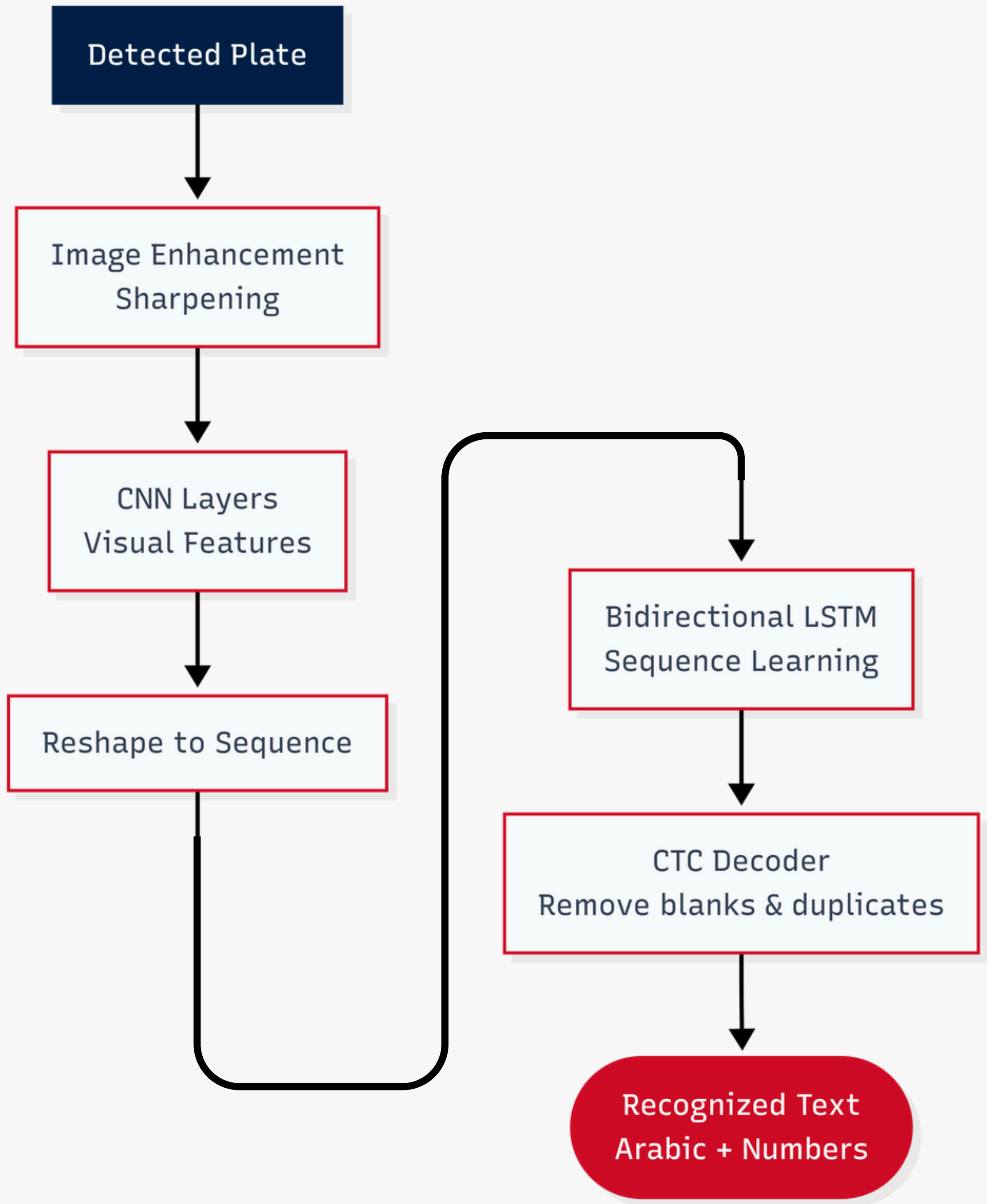
PERFORMANCE METRICS (GRAPHS)

INFERENCE SAMPLES

Technological Framework

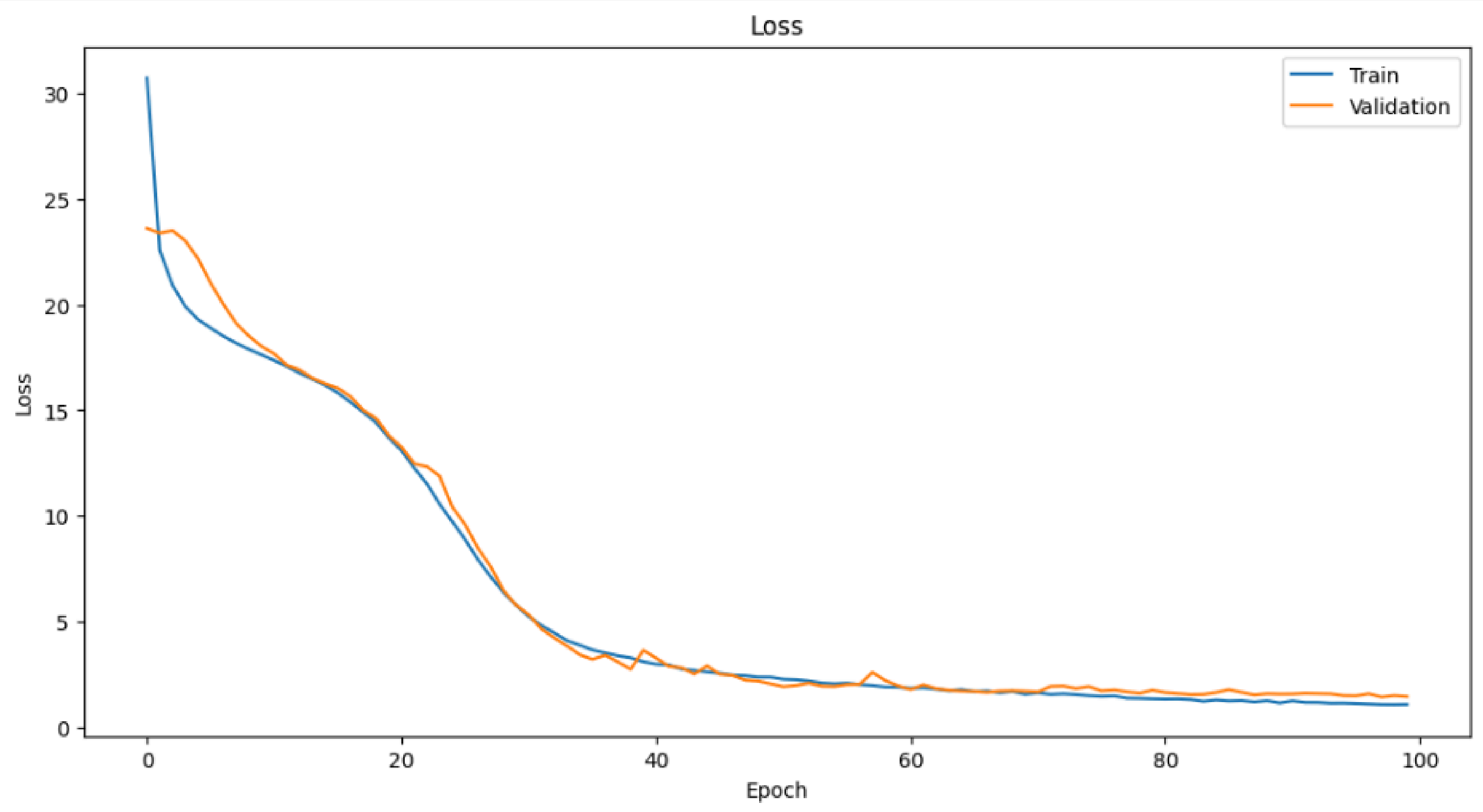
ARABIC OCR & SEQUENCE RECOGNITION PIPELINE

- **Plate ROI Input:** Processing the isolated plate image extracted from the detection phase.
- **Image Enhancement:** Applying Sharpening & Contrast filters to clarify complex Arabic scripts.
- **Feature Mapping (CNN):** Extracting deep visual patterns, focusing on character edges and spacing.
- **Sequence Conversion:** Transforming spatial features into a sequential data stream for time-series analysis.
- **Bidirectional LSTM (Bi-LSTM):** Analyzing text from both directions to understand context and font variations.
- **CTC Decoding:** Implementing Connectionist Temporal Classification for end-to-end alignment without manual segmentation.
- **Final Digital Output:** Delivering the recognized plate number as structured Bilingual text (Arabic & English).



Model Evaluation & Qualitative Results

ARABIC OCR & SEQUENCE RECOGNITION PIPELINE



PERFORMANCE METRICS (GRAPH)

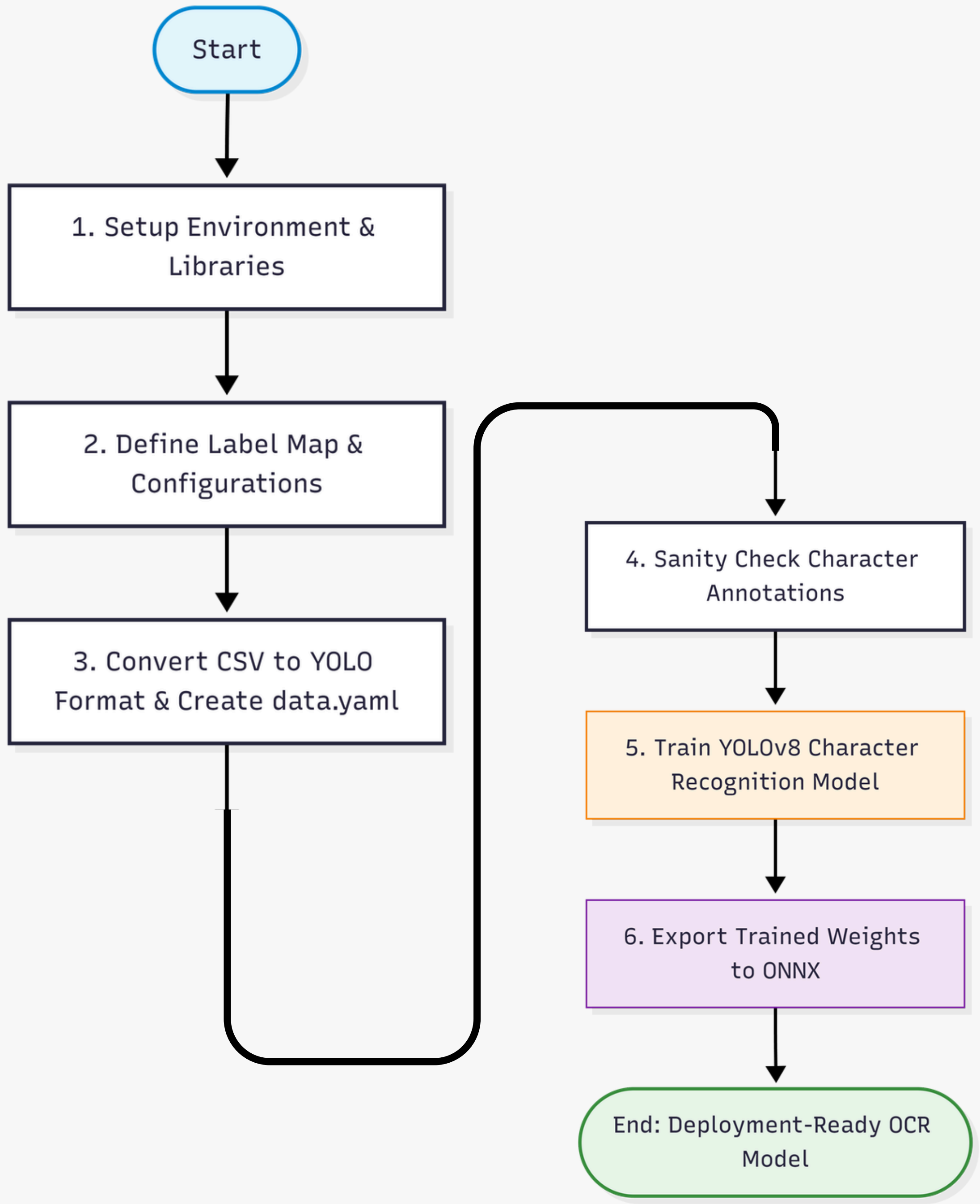

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 443ms/step
✅ Final Plate Text: ٧ ٤ ٦ ٢ ب ن س
```

INFERENCE SAMPLES

Technological Framework

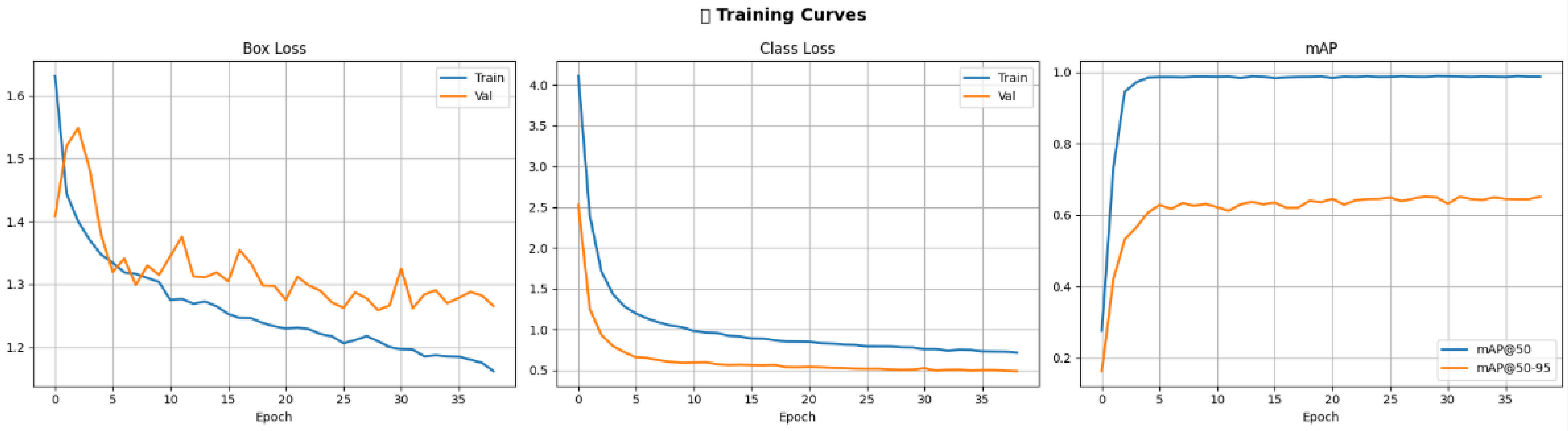
ARABIC OCR & SEQUENCE RECOGNITION PIPELINE - YOLO

- **Label Mapping & Config**: Map 38 unique Arabic character and digit classes and set training configurations.
- **YOLO Format Conversion:** Converted CSV pixel coordinates to normalized YOLO format and generated data.yaml.
- **Sanity Check:** Visualized sample training images to verify character bounding box alignment.
- **Model Training:** Fine-tuned the YOLOv8 model on custom car plate characters for 50 epochs.
- **Model Export:** Converted final trained weights to ONNX framework format for fast deployment.



Model Evaluation & Qualitative Results

ARABIC OCR & SEQUENCE RECOGNITION PIPELINE - YOLO



PERFORMANCE METRICS (GRAPHS)

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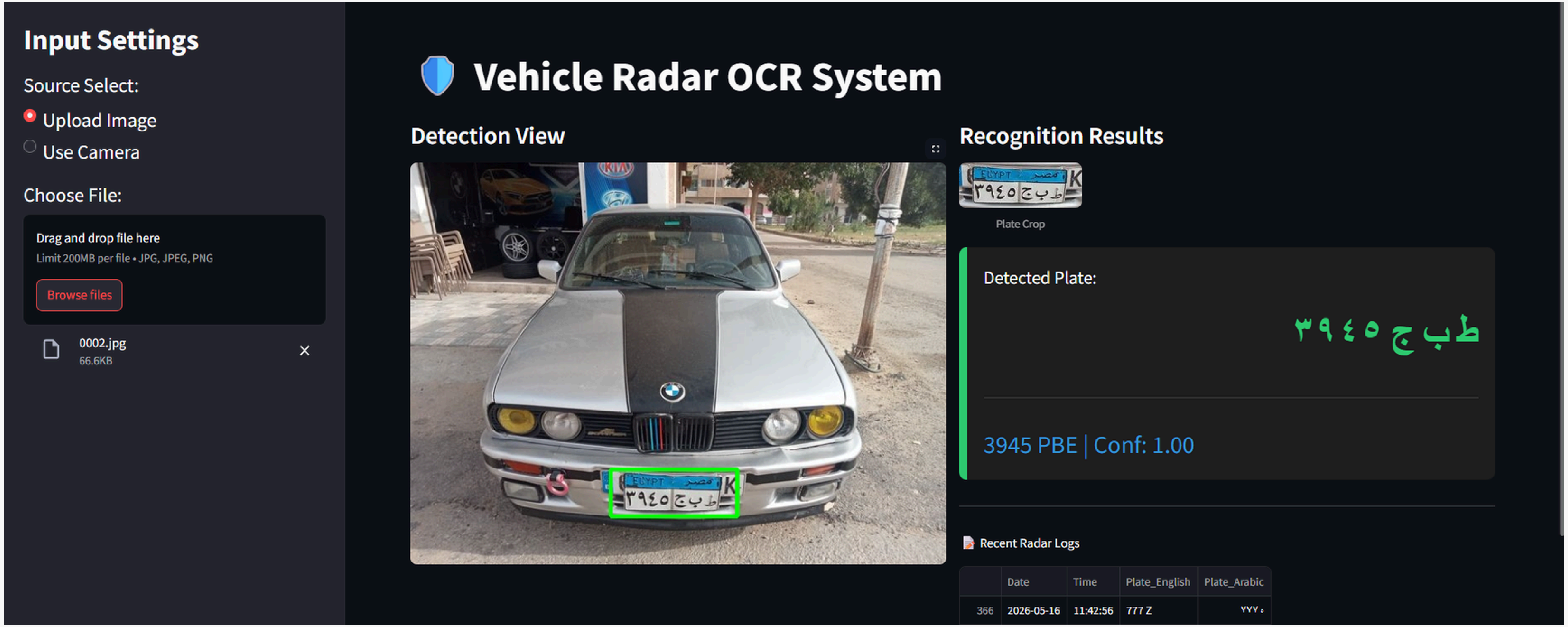
INFERENCE SAMPLES

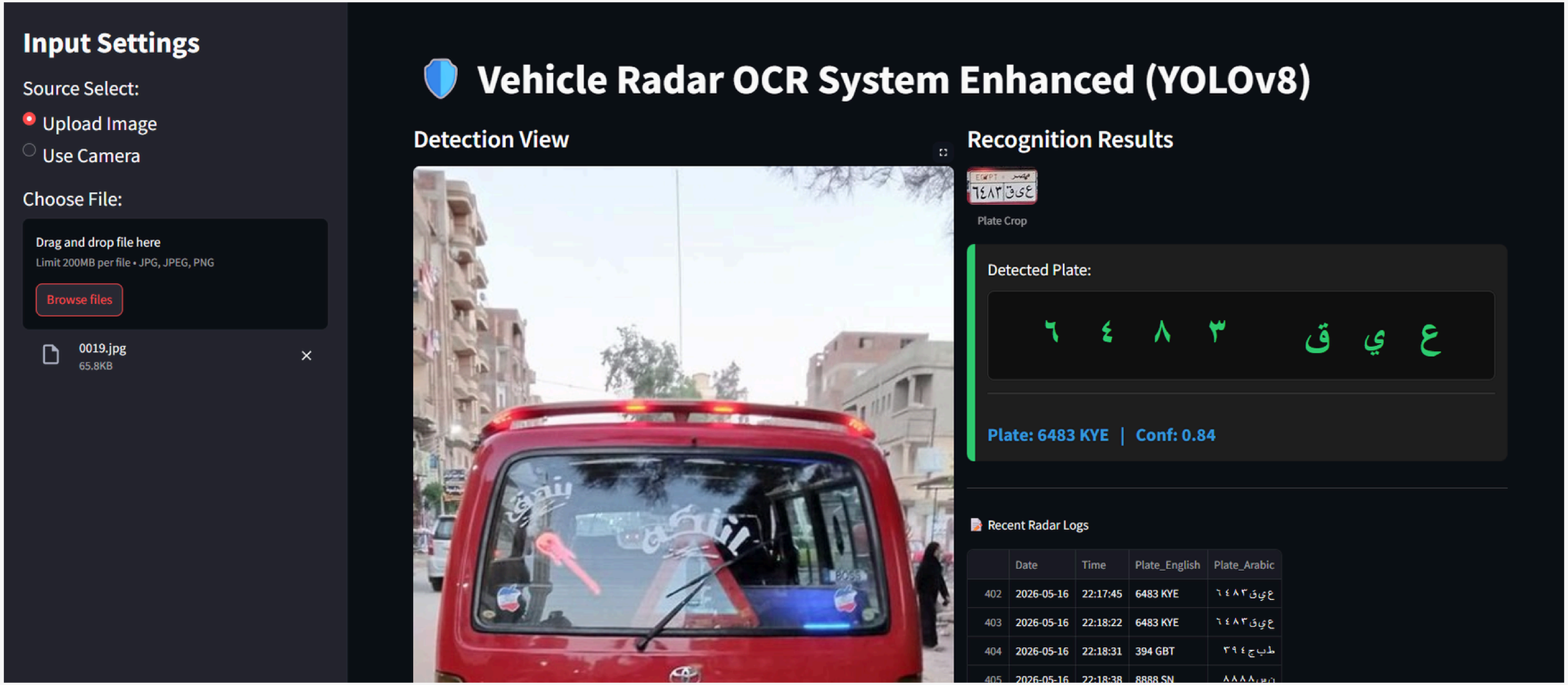
Deployment

CUSTOM CNN MODELS



Deployment

YOLO MODELS



Thank You